

Tutorial Problem Set VI

MATA30 – TUT0003

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During this tutorial, we discussed solutions to some problems on Quiz 2 and the midterm. Thereafter, we solved problems related to derivatives.

1 Basic problems

Problem 1. This question is based on errors that you, as a group, made on Quiz 2. Find counterexamples to the following statements. For (a) and (c), what additional assumptions can be made to make the statements true?

- (a) If $f, g : \mathbb{R} \rightarrow \mathbb{R}$ are functions such that $\lim_{x \rightarrow 0} (f(x) \cdot g(x)) = L$ for some $L \in \mathbb{R}$, then

$$\left(\lim_{x \rightarrow 0} f(x) \right) \cdot \left(\lim_{x \rightarrow 0} g(x) \right) = L.$$

- (b) For any $a, x \in \mathbb{R}$:

$$\sin(ax) = a \cdot \sin(x).$$

- (c) If $f, g : \mathbb{R} \rightarrow \mathbb{R}$ are functions such that $\lim_{x \rightarrow 0} g(x) = 0$, then $\lim_{x \rightarrow 0} f(x) \cdot g(x) = 0$.

Problem 2. In this exercise, you are asked to use the definition of a derivative.

- (a) Prove that the derivative of $f(x) = e^x$ is $f'(x) = e^x$. You may use without proof that

$$\lim_{h \rightarrow 0} \frac{e^h - 1}{h} = 1.$$

- (b) Prove that the derivative of $f(x) = \cos(x)$ is $f'(x) = -\sin(x)$.

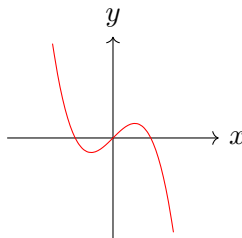
- (c) Prove that if $f, g : \mathbb{R} \rightarrow \mathbb{R}$ are differentiable functions, then $h(x) = f(x) + g(x)$ is differentiable with derivative $h'(x) = f'(x) + g'(x)$.

Problem 3. Compute the following derivatives.

- (a) Compute the derivative of $f(x) = \frac{\sin(x)}{x^2+1}$.

- (b) Compute the 100th derivative of the function $f(x) = xe^x$.

Problem 4. Suppose that $f : \mathbb{R} \rightarrow \mathbb{R}$ is a differentiable function such that $f(0) = 0$ and the graph of f' is given by:



- (a) Sketch the graph of f'' .
- (b) Sketch the graph of f .

Problem 5. Construct examples of the following.

- (a) A continuous but not differentiable function.
- (b) A differentiable function $f : \mathbb{R} \rightarrow \mathbb{R}$ such that f' is not differentiable.

Problem 6. Determine the lines through the point $(0, 2/3)$ that are tangent to the curve $y = x^2 - x^4$.

2 Challenging problems

Problem 7. Let $f : \mathbb{R} \rightarrow \mathbb{R}$ be a twice differentiable function. Show that

$$f''(x) = \lim_{h \rightarrow 0} \frac{f(x+h) + f(x-h) - 2f(x)}{h^2}.$$

Problem 8. In this problem, we will derive the *rocket equation* in physics. Consider a rocket travelling with velocity $v(t)$ at time t . Suppose that the rocket expells fuel at a constant rate of v_{ex} with respect to the rocket's frame and that the mass of the rocket together with the contained fuel is $m(t)$ at time t . Assume the following physical law.

Theorem 1 (Conservation of momentum). *Let P_i be particles of mass $m_i > 0$ and velocity $v_i(t)$ at time t with respect to some fixed coordinate system. Then, if there are no external forces,*

$$\sum_i m_i \cdot v'_i(t) = 0.$$

Prove that

$$m(t) \cdot v'(t) = m'(t) \cdot v_{\text{ex}}.$$